

Nathanael Tesfaye

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EDUCATION

Vanderbilt University, School of Engineering - Cumulative GPA: 3.61 Nashville, TN
Bachelor of Science in Computer Science and Mathematics Expected: May 2026
Relevant Coursework: Data Structures and Algorithms, Programming Languages, Operating Systems, Linear Algebra

TECHNICAL SKILLS

- **Languages:** Java, Python, Swift, C++, C, JavaScript, Kotlin, Racket, Prolog, Ruby, SQL, Bash
- **Frameworks/Libraries:** FastMCP, FastAPI, React.js, Angular, Next.js, CoreML, PyTorch, NumPy
- **Certifications:** Back-End Development and APIs w/ FreeCodeCamp, Software Engineering Simulation w/ JPMorgan

EXPERIENCE

Apple | Swift, React, Python, JavaScript, Docker, Kubernetes, TShark Cupertino, CA
Software Engineering Intern June 2025 – Sept 2025

- Built a scalable data ingestion pipeline leveraging MongoDB, AWS S3, and REST APIs, capable of storing and retrieving millions of network packets for large-scale analysis and long-term archival
- Designed and implemented a packet capture parsing service with a companion React-based web client, enabling engineers to visualize and diagnose Wi-Fi network issues five times faster than previous workflows
- Enhanced Apple's internal Network Testing application with AI-driven analysis and peer-to-peer connectivity, reducing troubleshooting time for 20,000+ internal users
- Built and optimized native iOS, macOS, and watchOS apps using UIKit, SwiftData, Combine, and Foundation Models

Trane Technologies | C, Ruby, Python, JavaScript, Selenium Tyler, TX
Software Engineering Intern May 2024 – Aug. 2024

- Created an application that allows employees to automatically upgrade and downgrade multiple indoor/outdoor units to any version simultaneously cutting time spent on manual labor in half
- Implemented automation scripts using Python to enhance and test system efficiencies in the "Technician App", reducing manual input time by more than 50% and improving system response times by 25%
- Researched and designed the integration of the next-generation user interface framework by analyzing alternatives to help save the company over \$1,000,000/annually in revenue

PROJECTS

Tic-Tac-Toe AI | Swift, CoreML, PyTorch, NumPy, AVFoundation, Core Data Github | View Demo

- Developed a Swift based Tic-Tac-Toe application allowing the user to play against a friend or the algorithm
- Utilized PyTorch to generate over 19,000 data points used for the [TicTacToeAI Model](#) via CreateML Tabular Classifier
- Implemented the MiniMax Algorithm to generate accurate data points and the RandomForest Algorithm for training

Postman Clone | Next.js, Google Auth, Tailwind CSS, MongoDB Github | View Project

- Developed a Postman-like API testing web application using Next.js, enabling users to view real-time API responses
- Optimized API response viewer to display formatted JSON/XML data, response metadata, and error handling
- Integrated MongoDB for dynamic storage of API requests and Google Auth for secure authentication/streamlined login

LEADERSHIP

National Society of Black Engineers Nashville, TN
Vice-President Apr. 2024 – Present

- Led the development and implementation of the chapter's strategic plan, resulting in a 25% increase in membership and a 15% increase in active member participation over the past academic year.
- Coordinated with university administration and external partners to secure \$20,000 in sponsorships and grants.

Google Developer Group Nashville, TN
Social Media Lead, Developer Sept. 2022 – Present

- Managed the Google Developer Student Group's Instagram account and GroupMe platform, delivering timely and engaging updates for all events, resulting in a 30% increase in member participation over eight months.